Theorem 6: upper bound for the edit distance

Correspondence between the paper and the Lean 4/Mathlib formalisation

Paper. J. Breitung, T. Gervens, M. Grohe. *Graph Similarity Meets Optimal Transport* (file `Graph_Similarity_Meets_Optimal_Transport.pdf` in the repository). Statement, equation, page and line numbers refer to this PDF; `docs/NUMBERING.md` lists the Lean files and declarations of all numbered statements.

The folder `GsimOT/Theorem6_EditUpperBound` formalises Theorem 6 (p. 7) with its proof in Appendix C.2 (Eq. (17) and the steps (a)–(g)) and the δ_{ed} part of Theorem 2. Shared notions (the edit distance, Eq. (3)) come from `GsimOT/Common`; they are listed at the end.

Checking. Run `bash build.sh` in the repository root (see `README.md`). The script builds the library and runs `scripts/CheckAxioms.lean`, which must report only the standard axioms `propext`, `Classical.choice`, `Quot.sound`: no `sorry` and no additional axioms.

Main statement in Lean. `GsimOT.theorem6`: for weights $\lambda_i \geq 0$ with $\sum_i \lambda_i = 1$ and bijections $\pi_i \colon V \to W$ over a finite index set, if $Q = \sum_i \lambda_i P_{\pi_i}$, then

$$\sum_i \lambda_i \, C_{\mathrm{ed}}(\pi_i) \leq 2 \, C_{\mathrm{ed}}(Q).$$

`GsimOT.theorem6_dEdit`: $d_{\mathrm{ed}}(G,H) \leq 2\, d_{\mathrm{ed}}^{\mathrm{OT}}(G,H)$, and `GsimOT.theorem2_edit` (Theorem 2, first bound): $\delta_{\mathrm{ed}}(G,H) \leq 2\, \delta_{\mathrm{ed}}^{\mathrm{OT}}(G,H)$, for all finite graphs G, H.

Correspondence: Theorem 6 and its proof

Paper	Page	Statement	Lean declaration(s) and location
Eq. (17) (l. 762–765)	24	$C_{\mathrm{ed}}(R) = \|A_G\|_F^2 + \|A_H\|_F^2 - 2\langle A_G R, R A_H\rangle_F$ for $R \in \mathrm{DS}(V,W)$, from $\|a-b\| = a^2 + b^2 - 2ab$ for $a, b \in \{0,1\}$	`GsimOT.costEdit_eq_frob` Theorem6_EditUpperBound/TraceFormula.lean:186 `GsimOT.abs_disc_eq` Theorem6_EditUpperBound/TraceFormula.lean:177 `GsimOT.frob` Theorem6_EditUpperBound/TraceFormula.lean:37 `GsimOT.adjMul` Theorem6_EditUpperBound/TraceFormula.lean:83 `GsimOT.mulAdj` Theorem6_EditUpperBound/TraceFormula.lean:86

Paper	Page	Statement	Lean declaration(s) and location
Proof of Thm. 6, first step (l. 767)	24	$C_{\mathrm{ed}}(\pi) = \|A_G - P_\pi A_H P_\pi^\top\|_F^2$	`GsimOT.costEditBij_eq_frob_conjMul` Theorem6_EditUpperBound/TraceFormula.lean:147 `GsimOT.conjMul` Theorem6_EditUpperBound/TraceFormula.lean:90 `GsimOT.conjMul_permMat` Theorem6_EditUpperBound/TraceFormula.lean:110
Proof of Thm. 6, step (a) (l. 768)	25	$\|A_G - P A_H P^\top\|_F = \|A_G P - P A_H\|_F$ (permutation invariance)	`GsimOT.frob_conjMul_eq` Theorem6_EditUpperBound/TraceFormula.lean:158 `GsimOT.adjMul_permMat` Theorem6_EditUpperBound/TraceFormula.lean:95 `GsimOT.mulAdj_permMat` Theorem6_EditUpperBound/TraceFormula.lean:103
Proof of Thm. 6, step (b) (l. 770)	25	$\|X + Y\|_F^2 \le 2\|X\|_F^2 + 2\|Y\|_F^2$	`GsimOT.frob_add_self_le` Theorem6_EditUpperBound/TraceFormula.lean:50
Proof of Thm. 6, steps (c), (f) (l. 771, 774)	25	$\|X - Y\|_F^2 = \|X\|_F^2 + \|Y\|_F^2 - 2\langle X, Y\rangle_F$	`GsimOT.frob_sub_self` Theorem6_EditUpperBound/TraceFormula.lean:44 `GsimOT.frob_comm` Theorem6_EditUpperBound/TraceFormula.lean:40
Proof of Thm. 6, steps (d), (g) (l. 772, 776)	25	$\mathbb{E}[P] = Q$: $A_G Q = \sum_i \lambda_i A_G P_{\pi_i}$, $Q A_H = \sum_i \lambda_i P_{\pi_i} A_H$, and $\langle\cdot,\cdot\rangle_F$ is linear; $\|A_G P_\pi\|_F = \|A_G\|_F$, $\|P_\pi A_H\|_F = \|A_H\|_F$	`GsimOT.adjMul_eq_sum` Theorem6_EditUpperBound/TraceFormula.lean:130 `GsimOT.mulAdj_eq_sum` Theorem6_EditUpperBound/TraceFormula.lean:138 `GsimOT.frob_eq_sum_left` Theorem6_EditUpperBound/TraceFormula.lean:56 `GsimOT.frob_eq_sum_right` Theorem6_EditUpperBound/TraceFormula.lean:68 `GsimOT.frob_adjMul_permMat` Theorem6_EditUpperBound/TraceFormula.lean:118 `GsimOT.frob_mulAdj_permMat` Theorem6_EditUpperBound/TraceFormula.lean:124
Theorem 6 (l. 245–246)	7, 24–25	$\sum_i \lambda_i C_{\mathrm{ed}}(\pi_i) \le 2\, C_{\mathrm{ed}}(Q)$ for $Q = \sum_i \lambda_i P_{\pi_i}$; the proof is the chain (a)–(g) and Eq. (17) of Appendix C.2 (l. 767–777)	`GsimOT.theorem6` Theorem6_EditUpperBound/Theorem6.lean:44

Paper	Page	Statement	Lean declaration(s) and location
Theorem 6 (“in particular”, l. 247, 783–785)	7, 25	$d_{\mathrm{ed}}(G,H) \le 2\, d_{\mathrm{ed}}^{\mathrm{OT}}(G,H)$ (in Lean for all G, H)	`GsimOT.theorem6_dEdit` Theorem6_EditUpperBound/Theorem6.lean:112
Theorem 2 (first bound, l. 119)	3	$\delta_{\mathrm{ed}}(G,H) \le 2\, \delta_{\mathrm{ed}}^{\mathrm{OT}}(G,H)$	`GsimOT.theorem2_edit` Theorem6_EditUpperBound/Theorem6.lean:129

Shared notions used (from `GsimOT/Common`)

Paper	Page	Statement	Lean declaration(s) and location
Section 1, Section 2	2, 5	$C_{\mathrm{ed}}(Q) = \sum_{v,w,v',w'} \lvert D(v,w,v',w')\rvert\, Q(v,w)\, Q(v',w')$ and $C_{\mathrm{ed}}(\pi) = C_{\mathrm{ed}}(P_\pi) = \lVert A_G^{\pi} - A_H\rVert_{\mathrm{ed}}$	`GsimOT.costEdit` Common/EditDistance.lean:35 `GsimOT.costEditBij` Common/EditDistance.lean:39 `GsimOT.costEditBij_eq` Common/EditDistance.lean:60
Section 1, Eq. (1), Section 2, Appendix A	2, 5, 14–15	$d_{\mathrm{ed}} = \min_\pi C_{\mathrm{ed}}(\pi)$ and $d_{\mathrm{ed}}^{\mathrm{OT}} = \min_{Q \in \mathrm{DS}(V,W)} C_{\mathrm{ed}}(Q)$ (infimum, attained); $d_{\mathrm{ed}}^{\mathrm{OT}} \le d_{\mathrm{ed}}$	`GsimOT.dEdit` Common/EditDistance.lean:42 `GsimOT.dOTEdit` Common/EditDistance.lean:45 `GsimOT.exists_dOTEdit_eq` Common/EditDistance.lean:109 `GsimOT.dOTEdit_le_dEdit` Common/EditDistance.lean:102
Section 1, Section 2	1–2, 5	$\delta_{\mathrm{ed}} = n^{-2} d_{\mathrm{ed}}$, $\delta_{\mathrm{ed}}^{\mathrm{OT}} = n^{-2} d_{\mathrm{ed}}^{\mathrm{OT}}$ (scaling by $1/n^2$)	`GsimOT.deltaEdit` Common/EditDistance.lean:48 `GsimOT.deltaOTEdit` Common/EditDistance.lean:51
Section 1, Section 2, Appendix A	2, 5, 14	$C_{\mathrm{ed}}(Q) = \lVert B(G,H,Q)\rVert_{\mathrm{ed}}$ for $Q \ge 0$ (entrywise 1-norm $\lVert\cdot\rVert_{\mathrm{ed}} = \lVert\cdot\rVert_{(1)}$)	`GsimOT.entrywiseNorm` Common/MatrixNorms.lean:53 `GsimOT.costEdit_eq_entrywiseNorm_B` Common/MatrixNorms.lean:119

Paper	Page	Statement	Lean declaration(s) and location
Section 2, Eq. (3)	5	Birkhoff–von Neumann: $Q = \sum_i \lambda_i P_{\pi_i}$ with $\lambda_i > 0$, $\sum_i \lambda_i = 1$ (also: over all $\sigma \in \mathrm{Perm}(V)$ with $\lambda \geq 0$); conversely every such combination is in $\mathrm{DS}(V, W)$	`GsimOT.exists_birkhoff_decomposition_pos` `Common/BirkhoffDecomposition.lean:72` `GsimOT.exists_birkhoff_decomposition` `Common/BirkhoffDecomposition.lean:42` `GsimOT.isDS_of_decomposition` `Common/BirkhoffDecomposition.lean:23`

Notes on the formalisation

1. The formalisation follows the proof of Theorem 6 in Appendix C.2. The expectation $\mathbb{E}[f(\pi)]$ over the random bijection with $\Pr(\pi = \pi_i) = \lambda_i$ is the sum $\sum_i \lambda_i f(\pi_i)$, and the chain of l. 767–777 is a `calc` block with one step per line, commented with the paper's line numbers and labels (a)–(g).

2. Matrices are functions $V \to W \to \mathbb{R}$, as in `Common/Basic.lean`: `adjMul G R` is $A_G R$, `mulAdj H R` is $R A_H$, `conjMul H R` is $R A_H R^\top$ and `frob X Y` is $\langle X, Y \rangle_F$, so $\|X\|_F^2$ is `frob X X`.

3. A decomposition is given by weights $\lambda_i \geq 0$ with $\sum_i \lambda_i = 1$ over any finite index type; only these are assumed, so `theorem6` applies to every decomposition as in Eq. (3).

4. The first step of the chain (l. 767) holds since the entries of A_G and A_H are in $\{0, 1\}$ (`costEditBij_eq_frob_conjMul`).

5. Example 7 (p. 7) and the remark at the end of Appendix C.2 (p. 25, arbitrary matrices via thresholds) are not formalised.